

Network discovery and traffic analysis

Practical lab report

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1). Introduction.

This practical lab was conducted to understand basic network discovery and traffic analysis in an authorized virtual lab environment. Nmap was used to perform targeted TCP and UDP scans against the designated target machine. The purpose of the scans was to identify open ports and determine the services running on the target.

Wireshark was then used to capture and examine network traffic between the Kali Linux scanning machine and the target system. The packet capture helped demonstrate how network communication can be observed and analyzed.

All scanning activities in this exercise were performed against an authorized local virtual machine in an isolated lab environment.

2). Lab environment

Component	Details
Attacker/Scanner	Kali Linux
Target	Metasploitable
Network	Host-only virtual network
Scanner tool	Nmap
Packet analysis tool	Wireshark
Target IP	192.168.31.4
Purpose	Authorized network discovery

The practical exercise was performed using two virtual machines connected through an isolated host-only virtual network. Kali Linux was used as the scanning machine, while Metasploitable was used as the authorized target.

Nmap was used for TCP and UDP port scanning and service identification. Wireshark was used to capture and analyze network traffic generated during the testing process.

3). Objectives.

The objectives of this practical exercise were:

1. To perform targeted TCP port scanning using Nmap.
2. To perform a small authorized UDP port scan.
3. To identify at least three services running on the target system.
4. To capture network traffic using Wireshark.
5. To analyze and document a relevant packet flow.
6. To present the findings in a professional cybersecurity report.

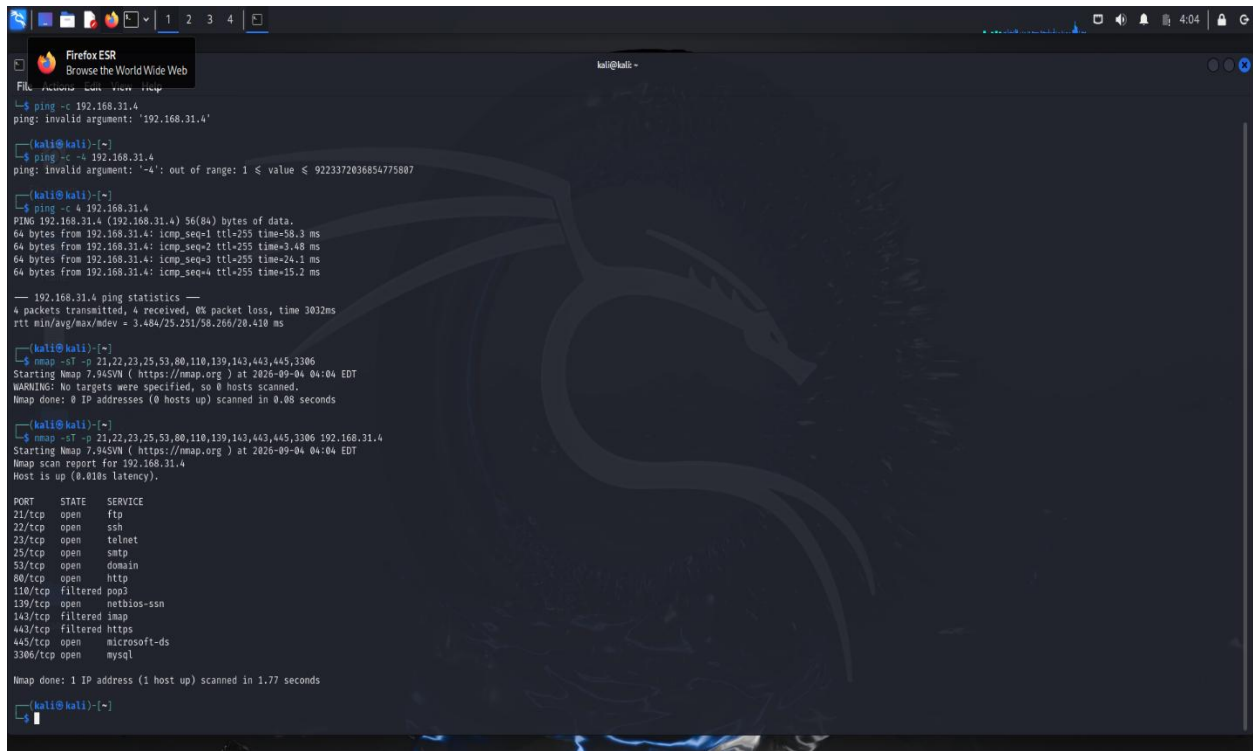
4). Methodology.

The target IP address was first identified from the authorized Metasploitable virtual machine. Connectivity between the Kali Linux machine and the target was then verified using ICMP ping.

A targeted TCP scan was performed using Nmap against selected commonly used ports. Service detection was subsequently performed to obtain additional information about the identified services.

A small UDP scan was then conducted against a limited selection of UDP ports. Finally, Wireshark was used to capture network traffic generated during communication with the target. The captured packets were filtered and examined to identify a relevant packet flow.

5). TCP scan.



```
kali@kali:~$ ping -c 4 192.168.31.4
ping: invalid argument: '192.168.31.4'

kali@kali:~$ ping -c -4 192.168.31.4
ping: invalid argument: '-4': out of range: 1 ≤ value ≤ 9223372036854775807

kali@kali:~$ ping -c 4 192.168.31.4
PING 192.168.31.4 (192.168.31.4) 56(84) bytes of data:
64 bytes from 192.168.31.4: icmp_seq=1 ttl=255 time=58.3 ms
64 bytes from 192.168.31.4: icmp_seq=2 ttl=255 time=3.48 ms
64 bytes from 192.168.31.4: icmp_seq=3 ttl=255 time=24.1 ms
64 bytes from 192.168.31.4: icmp_seq=4 ttl=255 time=15.2 ms

--- 192.168.31.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3032ms
rtt min/avg/max/mdev = 3.484/25.251/58.266/20.410 ms

kali@kali:~$ nmap -sT -p 21,22,23,25,53,80,110,139,143,443,445,3306
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-04 04:04 EDT
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.08 seconds

kali@kali:~$ nmap -sT -p 21,22,23,25,53,80,110,139,143,443,445,3306 192.168.31.4
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-04 04:04 EDT
Nmap scan report for 192.168.31.4
Host is up (0.818s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
110/tcp   filtered pop3
139/tcp   open  netbios-ssn
143/tcp   filtered imap
443/tcp   filtered https
445/tcp   open  microsoft-ds
3306/tcp  open  mysql

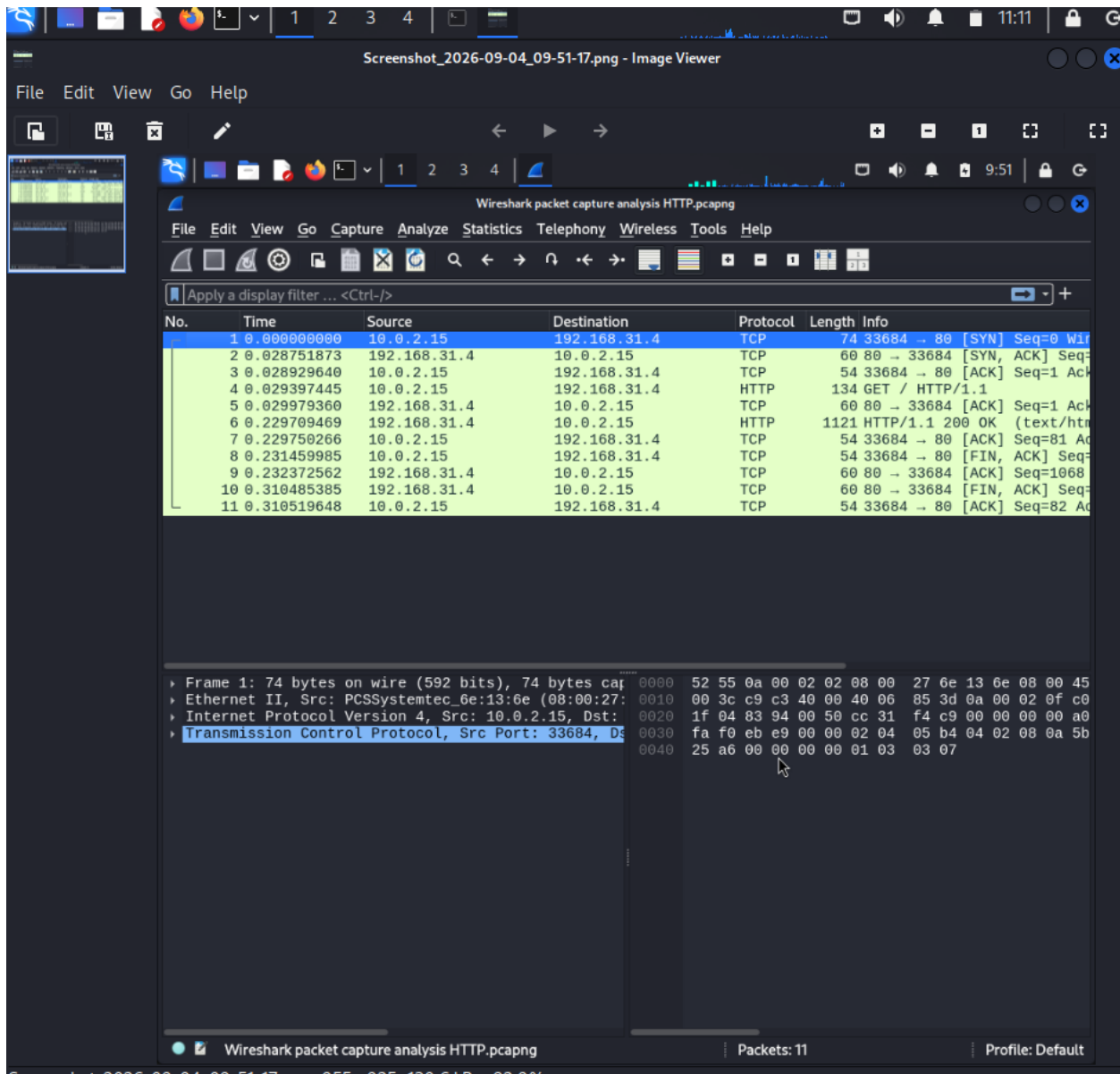
Nmap done: 1 IP address (1 host up) scanned in 1.77 seconds

kali@kali:~$
```

The -sT command is used to find TCP scan for open ports and check whether they are accessible. The -p command is used to find specific open ports in the Nmap scan.

The targeted TCP scan was performed using Nmap against the authorized Metasploitable target. The scan checked selected TCP ports commonly associated with network services. The results showed several open ports, indicating that corresponding services were accessible on the target system.

6). Wireshark scan.



Wireshark was used to capture network traffic between the Kali Linux system and the authorized target. Since Nmap identified TCP port 80 as an open HTTP service, HTTP traffic was generated using the `curl` command.

Wireshark was used to capture traffic between the Kali Linux system and the authorized target at 192.168.31.4. The `eth0` interface was selected because the routing table indicated that traffic to the target was being sent through this interface. HTTP traffic was generated using the `curl` command and the captured packets were filtered using `tcp.port == 80`.

7). Findings and Analysis.

The Nmap scans identified multiple open TCP services on the authorized target. Three of the identified services were FTP, SSH, and HTTP. The presence of these services indicates that the target provides several network-accessible functions.

The UDP scan also identified the state of selected UDP ports. UDP scanning produced different results from TCP scanning because UDP does not establish a connection in the same way as TCP.

The Wireshark capture provided additional evidence of network communication with the target. The ICMP Echo Request and Echo Reply packets demonstrated successful communication between the two virtual machines.

Overall, the exercise demonstrated the relationship between network discovery and packet analysis. Nmap provided information about accessible services, while Wireshark provided visibility into network traffic.

8). Conclusion.

This practical exercise demonstrated the basic process of network discovery and traffic analysis in an authorized virtual lab environment. Targeted TCP and UDP scans were performed using Nmap, and multiple services were identified on the target system.

Wireshark was used to capture and analyze network traffic, providing a practical understanding of how packets travel between systems. The exercise improved understanding of port scanning, service identification, and basic packet analysis.